Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		**Any 3 × (1) from:** – Volume of molecules is negligible compared to the size of the gas **or** time between collisions much longer than during – Collisions between the molecules are elastic [on average] – Forces of attraction between molecules are negligible / forces only applied when molecules collide / constant velocity between collisions / molecules have KE but no PE – Gases consist a [large number] of molecules in [rapid], random motion	3			3		
	(b)	(i)	$\text{rms speed} = \text{sqrt}\left(\frac{480^2+521^2+436^2+445^2+503^2}{5}\right)$(1) $= 478\ [\text{m s}^{-1}]$ (1)		2		2	2	
		(ii)	$m = \frac{32\times10^{-3}}{6.02\times10^{23}} = 5.316\times10^{-26}$ kg **or** 32 u (1) Use $\frac{1}{2}m\overline{c^2} = \frac{3}{2}kT$ or equivalent and rearrange (1) $T = \left(\frac{m}{3k}\right)\times \text{rms speed}^2 = \left(\frac{5.316\times10^{-26}}{3(1.38\times10^{-23})}\right)\times 478^2 = 293.4$ [K] (1) **Alternative:** Use $\frac{1}{2}M_r\overline{c^2} = \frac{3}{2}RT$ (1) Rearrange (1) Answer = 293.4 [K] (1)		3		3	3	
	(c)		Method for temperature variation e.g. the $(\text{rms speed})^2 \propto T$ (1) Answer i.e. $\times\sqrt{2}$ + comment e.g. Gareth is incorrect (1) Method for pressure variation e.g. $PV = nRT$ (1) Answer + comment e.g. Gareth is correct (1) Accept $p \propto T$ Gareth is correct for 1 mark only			4	4		
			Question 7 total	**3**	**5**	**4**	**12**	**5**	**0**